

## Lecture Unit 7.3

# Interconnect and messages

# Performance degradation

SAB: section 1.4.5

- A number of factors can contribute to real-world parallel performance being less than the idealised peak performance:
  - **S**tarvation: insufficient parallel work to keep the processors busy (insufficient parallel work, uneven distribution of work)
  - **L**atency: time taken for information to travel from one part of the system to another (e.g. accessing memory, **sending messages**)
  - **O**verhead: work required in addition to the computation (e.g. starting and stopping OpenMP parallel regions)
  - **W**aiting: multiple threads/processes accessing a shared resource (e.g. contention for memory or network bandwidth)

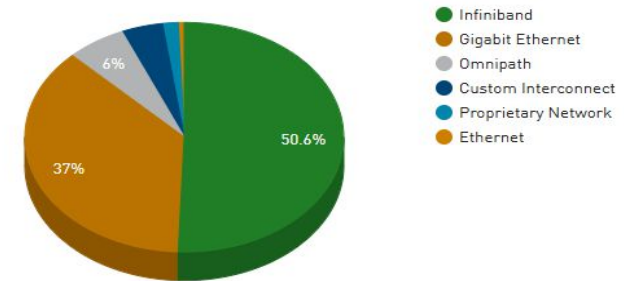
# Interconnect

- An MPI program has to send messages to communicate and co-ordinate between processes
- Compute nodes are linked by an interconnect(network) which carries these messages
- Time spent communicating needs to be minimised to achieve high parallel efficiency and scale to large numbers of processors
- We want to understand the impact on parallel performance

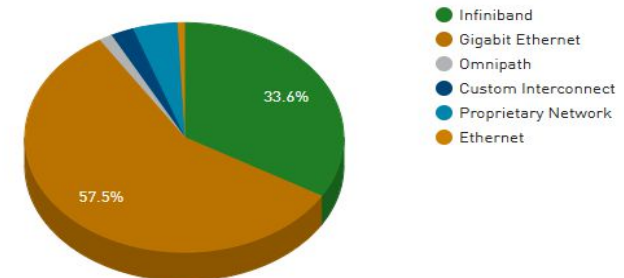
# Types of interconnect

- Gigabit ethernet and Infiniband are currently popular in clusters (e.g. Isca uses Infiniband)
- Significant part of the cost of an HPC system so may choose a cheaper interconnect if the performance of our workload is not dominated by communication

Interconnect Family System Share



Interconnect Family Performance Share

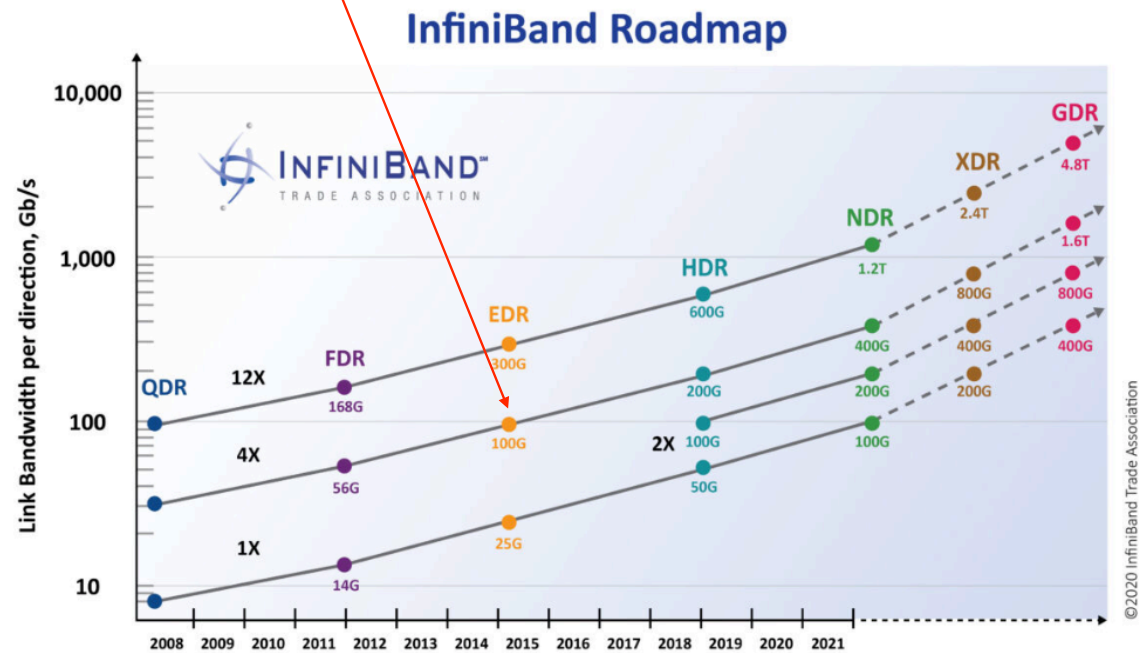


November 2024 list

<http://www.top500.org>

# Band Roadmap

Isca: 4xEDR (4 links aggregated)

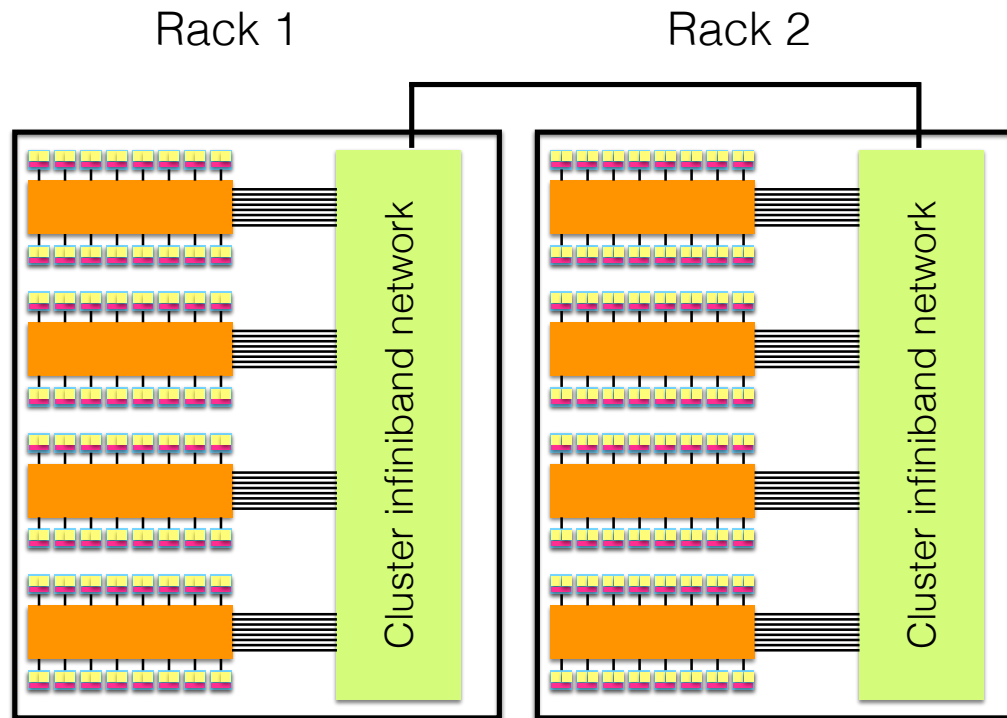


<https://www.infinibandta.org/infiniband-roadmap/>

# Network topologies

- Number of “hops” between nodes affects latency
- Connecting every node to every other node (fully connected) would be ideal but is only practical for a small network
- A bus or ring topology is simple but does not provide enough connectivity for HPC
- In practice more sophisticated topologies are used which balance cost and performance

# Cluster architecture

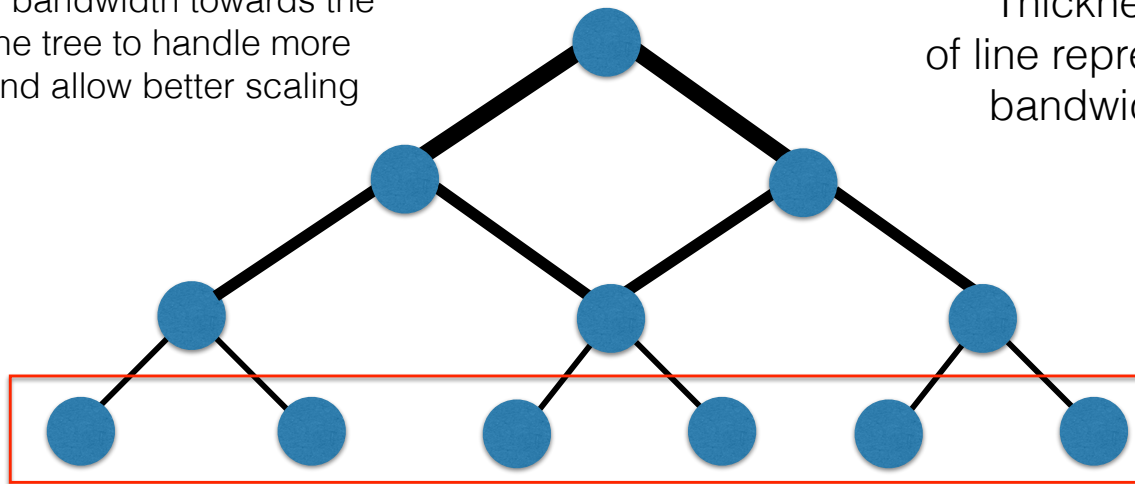


Connecting chassis in a way which scales needs care. This is a simplification.

# Fat tree

Greater bandwidth towards the top of the tree to handle more traffic and allow better scaling

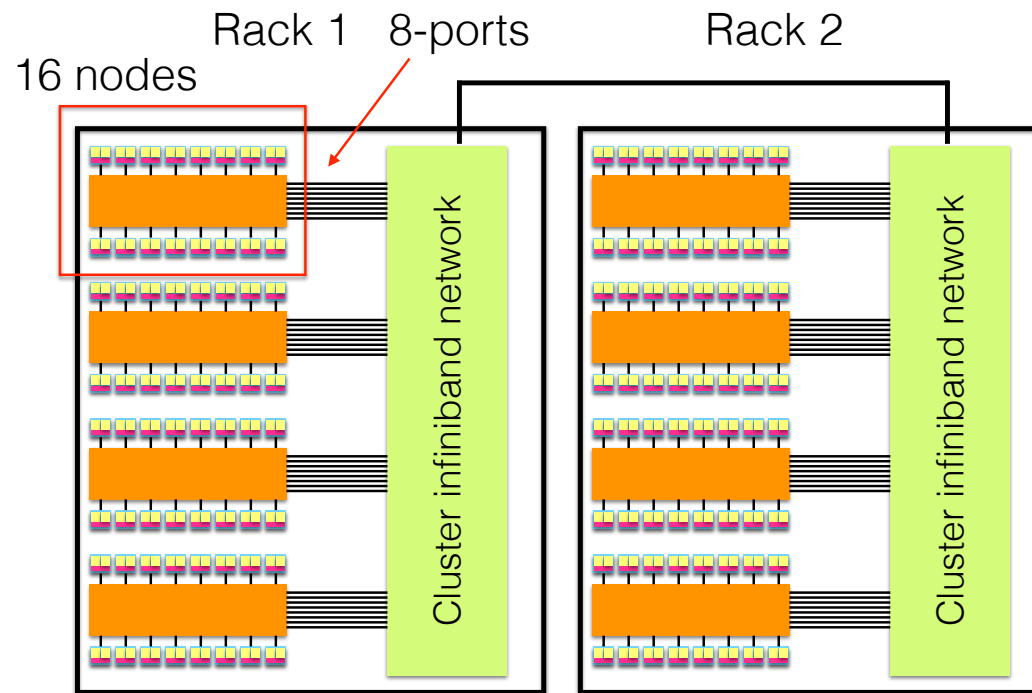
Thickness of line represents bandwidth



Leaves are compute nodes



# Cluster architecture



2:1 blocking network: reduced bandwidth when nodes are on different network switches

# MPI process placement

- Communication overhead between MPI processes depends on where the processes are running
  - MPI processes within the same compute node can communicate faster than process on different compute nodes
  - The speed of MPI communication between processes on different compute nodes depends on the network connectivity between the nodes
- The placement of MPI processes can be optimised to reduce communication time ([this paper](#) for an example of domain decomposition rank re-ordering)

# Modelling communications

- In practice the time taken to send a message from one node to
  - number of hops between nodes
  - blocking factor of the network
  - other network traffic
- We can model the best case where the transmission time depends only on a fixed latency (set up time) and a constant bandwidth. The transmission time is given by

$$t = L + \frac{M}{B}$$

where  $L$  is the latency,  $M$  is the message length and  $B$  is the bandwidth

# Transmission time: 1kB message

$$t = L + \frac{M}{B}$$

- Latency:  $L = 1\mu\text{s} = 1 \times 10^{-6}\text{s}$
- Bandwidth:  
 $B = 100 \text{ Gbit/s} = 12.5 \text{ GB/s} = 12.5 \times 10^9 \text{ bytes/s}$
- Message size:  $M = 1 \text{ kB} = 1000 \text{ bytes}$

$$t = 1 \times 10^{-6} + \frac{1000}{12.5 \times 10^9} = 1.08 \times 10^{-6} \text{ s}$$

- Transmission time is dominated by latency

# Transmission time: 1 MB message

$$t = L + \frac{M}{B}$$

- Latency:  $L = 1\mu\text{s} = 1 \times 10^{-6}\text{s}$
- Bandwidth:  
 $B = 100 \text{ Gbit/s} = 12.5 \text{ GB/s} = 12.5 \times 10^9 \text{ bytes/s}$
- Message size:  $M = 1 \text{ MB} = 1 \times 10^6 \text{ bytes}$

$$t = 1 \times 10^{-6} + \frac{1 \times 10^6}{12.5 \times 10^9} = 8.1 \times 10^{-5} \text{ s}$$

- Bandwidth is the more important factor

# Latency and bandwidth

- A high performance interconnect has:
  - Low latency: short delay sending small messages
  - High bandwidth: high data transfer rate for large messages
- Which one is more important depends on characteristics of the algorithm e.g. many small point-to-point communications vs. large collective communications
- May be able to aggregate smaller messages to reduce latency

# Unit Summary

- MPI programs need to send messages and we want to minimise communication time
- Infiniband and ethernet are networking technologies used in commodity clusters
- In practice a number of factors affect the transmission time for a message (number of hops, blocking factor, other traffic)
- We can model the best case transmission time as a fixed latency plus a constant bandwidth
- Latency matters more for small messages; bandwidth matters more for larger messages